

UPNM

National Defence University of Malaysia

Kewajipan ● Maruah ● Integriti

TPD3743

COMPUTER GRAPHICS AND VISUALISATION

Group Assignment

Animated Graphics Lab Assignment

STUDENT NAME	Student ID	TOTAL (20)
1. MAHESH A/L SIVAKANDAN	2240313	
2. MAGENDREN A/L PARTHIBAN	2240312	

CG GROUP ASSIGNMENT

Group Member's Details & Declaration of Contribution

We, the undersigned group members, declare that we have contributed to the completion of this assignment as detailed above. Each member has fulfilled their responsibilities, contributing to the successful completion of the project.

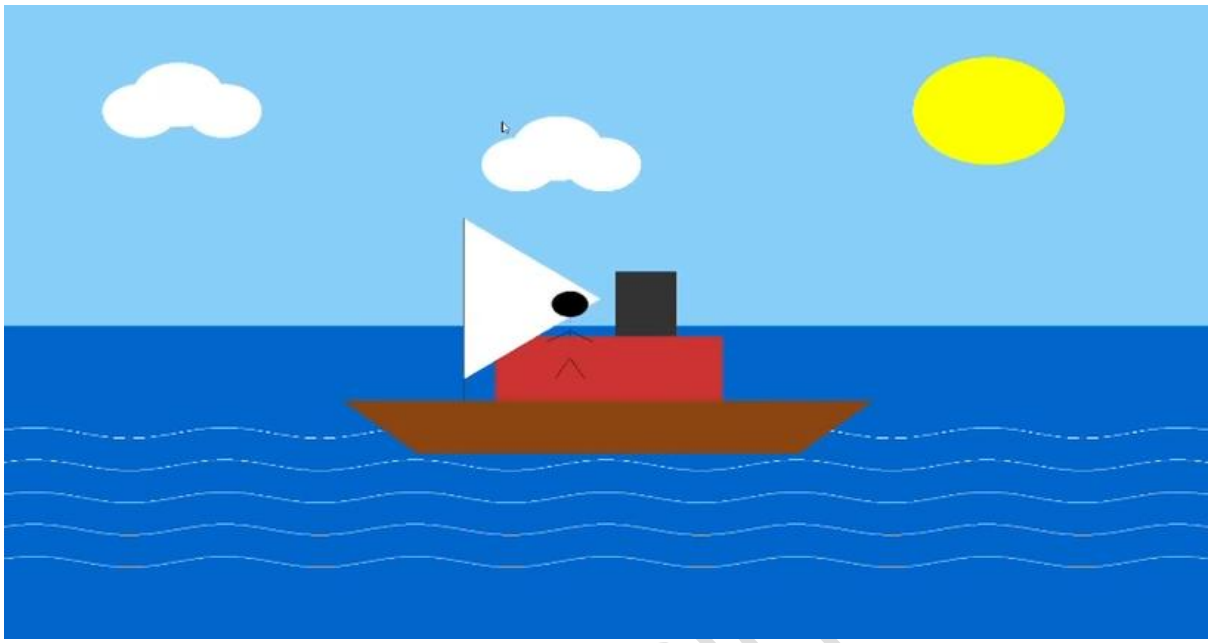
Student's Name	Student's ID	Signature	Date
MAHESH A/L SIVAKANDAN	2240313		17/6/2026
MAGENDREN A/L PARTHIBAN	2240312		17/6/2026

Animated Graphics Assignment Rubric (Total: 20 Marks)

Criteria	Needs Improvement (0-1 mark)	Good (2-3 marks)	Excellent (4-5 marks)	Marks Awarded
Environment Setup & Project Creation	Setup incomplete or project fails to build	Minor setup issues but project runs	Correctly installed tools, project created without errors	/5
Code Implementation (Animation Logic)	Animation incomplete or not functional	Basic animation works but lacks complexity	Smooth, well-structured animation with creative elements	/5
Recording & Submission (Youtube Video)	No video submitted or major issues	Video too short/long or visibility issue	15-30 sec clear recording, unlisted YouTube link provided	/3
Screenshot & Description	Missing screenshot or description	Screenshot or description lacking detail	High-quality screenshot + detailed paragraph explaining the animation	/4
Error Handling & Correctness	Major errors or crashes	Minor bugs but animation works	No crashes, handles edge cases, clean output	/3
Total Marks:				**/20**

CSV GROUP ASSIGNMENT

1. Screenshot of the animation



CGV GROUP ASSIG

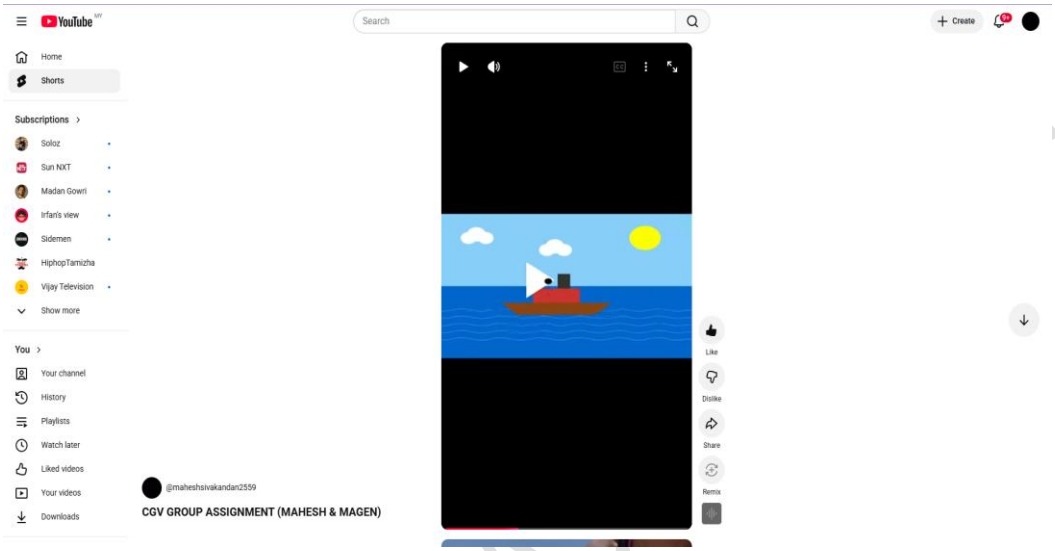
2. Description Paragraph

Our animation shows a ship sailing across the sea with moving waves, a bright sun, clouds in the sky, and a stickman standing on the ship. The animation was developed using Python in Visual Studio, where graphical objects were drawn and updated continuously through a loop to create the illusion of movement. The ship moves smoothly across the screen while the waves animate repeatedly, making the sea appear more realistic. Special features of the animation include the moving ship, animated ocean waves, sky elements such as the sun and clouds, and the stickman character that adds detail to the scene. One of the challenges we faced was making the ship move smoothly without glitches and ensuring that all animated objects remained synchronized during execution. We also had to adjust the positions and speeds of different objects so that the animation looked natural and visually appealing. Through this project, we learned how to use Python graphics programming, animation loops, object movement, and coordinate manipulation to create an interactive computer animation in Visual Studio.

CGV GROUP ASSIGNMENT

3. YouTube Video Link

https://youtube.com/shorts/uDtSbDn2JiE?feature=share



CGV GROUP AS

4. Project Python Code

```
#include <GL/glut.h>
#include <cmath>
float shipX = -600.0f;
float waveOffset = 0.0f;
float cloudX = 0.0f;
void drawCircle(float cx, float cy, float r)
{
    glBegin(GL_POLYGON);
    for (int i = 0; i < 360; i++)
    {
        float angle = i * 3.14159f / 180.0f;
        glVertex2f(cx + r * cos(angle),
            cy + r * sin(angle));
    }
    glEnd();
}
void drawCloud(float x, float y)
{
    glColor3f(1.0f, 1.0f, 1.0f);
    drawCircle(x, y, 25);
    drawCircle(x + 25, y + 15, 30);
    drawCircle(x + 55, y, 25);
}
void display()
{
    glClear(GL_COLOR_BUFFER_BIT);
    // SKY
    glColor3f(0.53f, 0.81f, 0.98f);
    glBegin(GL_QUADS);
    glVertex2f(0, 300);
    glVertex2f(800, 300);
    glVertex2f(800, 600);
    glVertex2f(0, 600);
    glEnd();
    // SEA
    glColor3f(0.0f, 0.4f, 0.8f);
    glBegin(GL_QUADS);
    glVertex2f(0, 0);
    glVertex2f(800, 0);
```

```
    glVertex2f(800, 300);
    glVertex2f(0, 300);
    glEnd();
    // SUN
    glColor3f(1.0f, 1.0f, 0.0f);
    drawCircle(650, 500, 50);
    // MOVING CLOUDS
    drawCloud(100 + cloudX, 500);
    drawCloud(350 + cloudX, 450);
    // WAVES
    glColor3f(1.0f, 1.0f, 1.0f);
    for (int y = 80; y <= 220; y += 30)
    {
        glBegin(GL_LINE_STRIP);
        for (int x = 0; x <= 800; x++)
        {
            float waveY =
                y + 5 * sin((x + waveOffset) * 0.05);
            glVertex2f(x, waveY);
        }
        glEnd();
    }
    // SHIP
    glPushMatrix();
    glTra
    // Sh
    glCol
    glBegin(GL_POLYGON);
    glVertex2f(250, 180);
    glVertex2f(500, 180);
    glVertex2f(550, 230);
    glVertex2f(200, 230);
    glEnd();
    // Deck
    glColor3f(0.8f, 0.2f, 0.2f);
    glBegin(GL_QUADS);
    glVertex2f(300, 230);
    glVertex2f(450, 230);
    glVertex2f(450, 290);
```

void __stdcall glPushMatrix()
Search Online

```
79          glVertex2f(300, 290);
80          glEnd();
81          // Chimney
82          glColor3f(0.2f, 0.2f, 0.2f);
83          glBegin(GL_QUADS);
84          glVertex2f(380, 290);
85          glVertex2f(420, 290);
86          glVertex2f(420, 350);
87          glVertex2f(380, 350);
88          glEnd();
89          // Mast
90          glColor3f(0.4f, 0.2f, 0.0f);
91          glBegin(GL_LINES);
92          glVertex2f(280, 230);
93          glVertex2f(280, 400);
94          glEnd();
95          // Sail
96          glColor3f(1.0f, 1.0f, 1.0f);
97          glBegin(GL_TRIANGLES);
98          glVertex2f(280, 400);
99          glVertex2f(280, 250);
100         glVertex2f(370, 325);
101         glEnd();
102         // STICKMAN
103         // Head
104         glColor3f(0.0f, 0.0f, 0.0f);
105         drawCircle(350, 320, 12);
106         // Body + Arms + Legs
107         glBegin(GL_LINES);
108         // Body
109         glVertex2f(350, 308);
110         glVertex2f(350, 270);
111         // Left Arm
112         glVertex2f(350, 295);
113         glVertex2f(335, 285);
114         // Right Arm
115         glVertex2f(350, 295);
116         glVertex2f(365, 285);
117         // Left Leg
```

```
118         glVertex2f(350, 270);
119         glVertex2f(340, 250);
120         // Right Leg
121         glVertex2f(350, 270);
122         glVertex2f(360, 250);
123         glEnd();
124         glPopMatrix();
125         glFlush();
126     }
127     void timer(int value)
128     {
129         shipX += 1.5f;
130         waveOffset += 2.0f;
131         cloudX += 0.2f;
132         if (shipX > 850)
133             shipX = -600;
134         if (cloudX > 900)
135             cloudX = -500;
136         glutPostRedisplay();
137         glutTimerFunc(16, timer, 0);
138     }
139     void init()
140     {
141         glClearColor(0.8f, 0.9f, 1.0f, 1.0f);
142         glMatrixMode(GL_PROJECTION);
143         gluOrtho2D(0, 800, 0, 600);
144     }
145     int main(int argc, char** argv)
146     {
147         glutInit(&argc, argv);
148         glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
149         glutInitWindowSize(800, 600);
150         glutInitWindowPosition(100, 100);
151         glutCreateWindow("Ship Sailing on the Sea");
152         init();
153         glutDisplayFunc(display);
154         glutTimerFunc(0, timer, 0);
155         glutMainLoop();
156         return 0;
```